

Matrix product operators, and finding the ground state with DMRG

FYS4480/FYS9480 – Week 41, October 8–9, 2026

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Thursday: matrix product operators, what their bond dimension counts, and fermions

Friday: the DMRG — environments, the two-site update, the sweep

Reading: lecture notes chapter 11, sections 11.3 and 11.5–11.7; Schollwöck, Annals of Physics 326, 96 (2011),
sections 4 and 6.

Code: `BookPrograms/chapterdmrg/dmrg.py`; notebook `week41.ipynb`.

Plan for week 41

Thursday, 2×45 min: matrix product operators

- The MPO, and why its matrices are not $d \times d$
- Automata: Heisenberg $D = 5$, pairing $D = 4$ with every level coupled
- Where the nonzero entries sit — and what a *bulk* entry means
- How many channels must cross a cut: the operator Schmidt rank
- Fermions: Jordan–Wigner strings, and the three $\hat{Z}$'s of the Hubbard MPO

Friday, 2×45 min: the DMRG algorithm

- Variational principle on the MPS manifold; the norm matrix G
- Environments and the effective Hamiltonian; why the cost is χ^3
- The two-site update, and exactly how the centre moves
- The discarded weight as an error indicator; extrapolation
- Exercises: the week-41 set (MPS by hand, canonical forms, parameter counting) and answers to the week-40 set (stability matrix, positive semidefinite matrices)

Where we got to last week

$$|\Psi\rangle = \sum_{\{s\}} A^{s_1} \cdots A^{s_i} \Lambda_i B^{s_{i+1}} \cdots B^{s_L} |s_1 \cdots s_L\rangle$$

What canonical form buys

- $\sum_s A^{s\dagger} A^s = I$: left block states orthonormal
- $\sum_s B^s B^{s\dagger} = I$: right block states orthonormal
- Λ_i is the Schmidt spectrum of that cut

So, for free

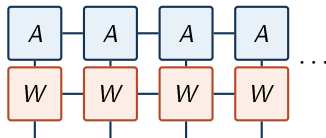
- $\langle\Psi|\Psi\rangle = \|\Lambda_i\|_F^2$
- $S_i = -\sum_a \sigma_a^2 \ln \sigma_a^2$
- truncation at bond i is optimal

We can **write** states. This week: how to write *operators*, and how to *find* the ground state.

Thursday: Matrix product operators

Matrix product operators

$$\hat{H} = \sum_{\{s\}, \{s'\}} W^{s_1 s'_1} W^{s_2 s'_2} \dots W^{s_L s'_L} |s_1 \dots s_L\rangle \langle s'_1 \dots s'_L|$$



- $\hat{H}|\Psi\rangle$ is again an MPS, with bond dimension χD
- *Automaton*: a machine walks the chain, placing operators, remembering what is open
- D counts the partial terms open at a cut – not the terms in $\hat{H}$

Why the matrices are not $d \times d$

Each W_i has *four* indices, and they do two different jobs:

$$W_{\alpha\beta}^{s s'}, \quad \underbrace{s, s' = 1, \dots, d}_{\text{physical: in and out}} \quad \underbrace{\alpha, \beta = 1, \dots, D}_{\text{bond: the automaton's state}}$$

- W is a $D \times D$ matrix whose entries are $d \times d$ operators
- The s, s' are contracted with the *state*; the α, β with the *neighbouring* W
- D has nothing to do with d : Heisenberg $d = 2, D = 5$; Hubbard $d = 4, D = 6$

The automaton's memory lives on the bond index. Shape: (D, D, d, d) .

The Heisenberg chain: $D = 5$

$$\hat{H} = J \sum_i S_i \cdot S_{i+1}, \quad W = \begin{pmatrix} \hat{1} & S^x & S^y & S^z & 0 \\ 0 & 0 & 0 & 0 & JS^x \\ 0 & 0 & 0 & 0 & JS^y \\ 0 & 0 & 0 & 0 & JS^z \\ 0 & 0 & 0 & 0 & \hat{1} \end{pmatrix}$$

First site: the first row. Last site: the last column.

- The five states: nothing placed; S^x , S^y or S^z waiting; finished
- Each path $1 \rightarrow \alpha \rightarrow 5$ places exactly one bond term
- An $L = 100$ chain has 297 terms and still $D = 5$

Pause to think: what the bond dimension counts

Pause to think #1

(about 4 min – discuss with your neighbour)

1. An $L = 100$ Heisenberg chain has 297 terms, and $D = 5$.
Explain the discrepancy in one sentence.
2. What is D for $J_1 \sum_i S_i \cdot S_{i+1} + J_2 \sum_i S_i \cdot S_{i+2}$?

Answer

Answer to pause #1

1. D counts terms open *at one cut*, and at most one bond term straddles a cut; the 297 terms are *paths* through the network, not its width.
2. $D = 8$: each component now needs two waiting states, $1 + 3 + 3 + 1$.

Where do the nonzero entries sit?

region of W	transition	meaning
first row, $W_{1\beta}$	idle $\rightarrow \beta$	start a term at this site
last column, $W_{\alpha D}$	$\alpha \rightarrow$ done	finish a term at this site
bulk, $W_{\alpha\beta}, 1 < \alpha, \beta < D$	$\alpha \rightarrow \beta$	carry a half-finished term across this site

- Nearest neighbour: the machine waits exactly *one* step, so the bulk is *empty*
- **Bulk entries are memory** – they appear when something must be remembered longer

Three bulk entries, three kinds of long range

Range 2

$$W = \begin{pmatrix} \hat{1} & \hat{o} & 0 & 0 \\ 0 & 0 & \hat{1} & 0 \\ 0 & 0 & 0 & \hat{o} \\ 0 & 0 & 0 & \hat{1} \end{pmatrix}$$

$D = r + 2$ at range r

Exponential

$$W = \begin{pmatrix} \hat{1} & \hat{o} & 0 \\ 0 & \lambda \hat{1} & \lambda \hat{o} \\ 0 & 0 & \hat{1} \end{pmatrix}$$

$\sum_{i < j} \lambda^{j-i} \hat{o}_i \hat{o}_j$, $D = 3$ at *any* range

$$\lambda(\hat{N} - N_0)^2$$

Same automaton, **no decay at all**:

$$W = \begin{pmatrix} \hat{1} & \hat{n} & * \\ 0 & \hat{1} & 2\lambda \hat{n} \\ 0 & 0 & \hat{1} \end{pmatrix}$$

$$* = \lambda(1 - 2N_0) \hat{n} + \frac{\lambda N_0^2}{L} \hat{1}$$

$\mathcal{O}(L^2)$ terms, **one bit** of memory.

A self-loop sums a geometric series for free. Added to an existing MPO, the penalty costs **one** extra state.

$\hat{o}$ is any Hermitian one-site operator; $\hat{n} = \hat{a}^\dagger \hat{a} = \text{diag}(0, 1)$ is the *local* occupation, $\hat{N} = \sum_p \hat{n}_p$ the *global* one, and N_0 a plain number.

Two different λ 's: a *decay factor* in the middle column, a *strength* in the right one.

How many channels must cross a cut?

Split $\hat{H}$ at a bond: $\hat{H} = \hat{H}_A \otimes \hat{1} + \hat{1} \otimes \hat{H}_B + \sum_k \hat{X}_k \otimes \hat{Y}_k$, and count.

$D_{\min}$ = operator Schmidt rank (the SVD of $\hat{H}$, with $\langle \hat{X}, \hat{Y} \rangle = \text{tr}(\hat{X}^\dagger \hat{Y})$)

	rank at each cut				
Heisenberg, open ($L = 6$)	4	5	5	5	4
Heisenberg, ring ($L = 6$)	4	8	8	8	4
pairing, open ($L = 4$)	4	4	4		

- A ring cuts *two* bonds: $1 + 1 + 3 + 3 = 8$. The operator is the cheap part — χ roughly squares
- The automaton gives a valid D ; only the rank gives *the* smallest

The staircase shape is a gauge, not a property of $\hat{H}$

$$W^{[i]} \longrightarrow X_{i-1}^{-1} W^{[i]} X_i \quad \text{leaves } \hat{H} \text{ and } D \text{ unchanged}$$

Measured

Random X on the $D = 5$ Heisenberg MPO: all 25 blocks become nonzero, and $\max |\Delta \hat{H}| = 1.2 \times 10^{-14}$.

- The readable upper-triangular form is the *automaton gauge*
- An SVD-compressed MPO is normally **dense** — not a bug
- The *sparsity pattern* is gauge-dependent; $\hat{H}$ itself, D and the operator-Schmidt data are not

Fermions on a chain: Jordan–Wigner

$$a_m^\dagger = \left(\prod_{k < m} \hat{Z}_k \right) \hat{\sigma}_m^+ \quad \Longrightarrow \quad a_i^\dagger \hat{a}_j = \hat{\sigma}_i^+ \underbrace{\hat{Z}_{i+1} \cdots \hat{Z}_{j-1}}_{\text{the string}} \hat{\sigma}_j^-$$

The string runs over the orbitals **strictly between** i and j .

- Nearest neighbours: *no string at all*, $a_i^\dagger \hat{a}_{i+1} = \hat{\sigma}_i^+ \hat{\sigma}_{i+1}^-$
- Pairing: strings **cancel** – $\hat{P}_p^\pm$ is *even*, both operators on one site
- Strings cost *no* bond dimension. They cost **locality**: on a ring or in 2D, length $\mathcal{O}(L)$

A trap to avoid

Test a fermionic MPO at range *two*, never at range one: at range one there is no intermediate site, so writing $\hat{1}$ for $\hat{Z}$ cannot show. At $r = 2$ it is off by $\max |\Delta \hat{H}| = 2t$.

The Hubbard MPO: three different $\hat{Z}$'s

	where	what it does
(i) within-site	inside $\hat{c}_{\downarrow}^{\dagger} = \hat{Z} \otimes \hat{\sigma}^{+}$	the down operator passes the up orbital of its <i>own</i> site
(ii) between-site	on other sites' orbitals	the genuine Jordan–Wigner string
(iii) $\hat{F} = \hat{Z} \otimes \hat{Z}$	a cell of W	parity of a <i>whole</i> site – what the automaton carries

orbital:	1 $\uparrow$ 1 $\downarrow$	2 $\uparrow$ 2 $\downarrow$	
	site 1	site 2	
$\hat{c}_{1\uparrow}^{\dagger} \hat{c}_{2\uparrow}$	$\hat{\sigma}^{+}$ $\hat{Z}$	$\hat{\sigma}^{-}$	string <i>inside site 1</i>
$\hat{c}_{1\downarrow}^{\dagger} \hat{c}_{2\downarrow}$		$\hat{Z}$ $\hat{\sigma}^{-}$	string <i>inside site 2</i>

The up hop jumps over its *own* site's down orbital; the down hop over the *next* site's up orbital.

One rule for both spins, and what $\hat{F}$ is really doing

$$\hat{c}_{i\sigma}^\dagger \hat{c}_{i+1,\sigma} = (\hat{c}_\sigma^\dagger \hat{F})_i \otimes (\hat{c}_\sigma)_{i+1}$$
$$\underbrace{\hat{c}_\uparrow^\dagger \hat{F} = \hat{\sigma}^+ \otimes \hat{Z}}_{\hat{F} \text{ adds the } \hat{Z}} \qquad \underbrace{\hat{c}_\downarrow^\dagger \hat{F} = \hat{1} \otimes \hat{\sigma}^+}_{\hat{F} \text{ cancels the } \hat{Z}}$$

- For $\downarrow$, the within-site $\hat{Z}$ was the parity of the *same* site's up orbital — not between the two operators, so it must go
- The $\hat{Z}$ the down hop *does* need reappears on the right, inside $\hat{c}_\downarrow = \hat{Z} \otimes \hat{\sigma}^-$
- Hubbard: $D = 6$ (ring: 10). Verified against bare fermions, $\max |\Delta \hat{H}| = 0$

A trap to avoid

The h.c. channel is $(\hat{F} \hat{c}_\sigma)_i \otimes (\hat{c}_\sigma^\dagger)_{i+1}$, *not* $\hat{c}_\sigma \hat{F}$: $\{\hat{c}, \hat{F}\} = 0$. Writing it the wrong way gives $\max |\Delta \hat{H}| = 2t$ **and exactly the right ground-state energy**. A correct energy is *not* a test of an MPO.

Bond dimensions to know

From the automaton

- Ising $D = 3$; Heisenberg $D = 5$
- Pairing $D = 4$; $+\lambda(\hat{N} - N_0)^2$: $D = 5$
- Hubbard chain $D = 6$ (ring 10)
- Particle-hole term $D = 14$
- Range- r hopping, spinless: $D = 2r + 2$

Friday: The DMRG algorithm

The variational principle, on a new manifold

$$E[\Psi] = \frac{\langle \Psi | \hat{H} | \Psi \rangle}{\langle \Psi | \Psi \rangle} \quad \text{minimised over MPS with bond dimension } \leq \chi$$

- The trial space is no longer a set of determinants: it is **states with bounded entanglement**
- An MPS is a *product*, so with all but one tensor frozen it is **linear** in the last one
- So each local problem is an ordinary eigenproblem — and we **sweep**

Many-body connection

Exactly the structure of the self-consistent field: a hard non-linear problem solved by iterating easy linear ones. The difference: each local step minimises over its subspace — exactly in the ideal algorithm, in practice up to the eigensolver tolerance.

Being variational has a price: the norm matrix G

We want a **variational** method, so we minimise a *ratio* — and the denominator is where G is born. With everything frozen except Θ , write $|\Psi\rangle = \sum_{\alpha} \Theta_{\alpha} |\phi_{\alpha}\rangle$:

$$E = \frac{\langle \Psi | \hat{H} | \Psi \rangle}{\langle \Psi | \Psi \rangle} = \frac{\Theta^{\dagger} \hat{H}_{\text{eff}} \Theta}{\Theta^{\dagger} G \Theta}, \quad G_{\alpha\beta} = \langle \phi_{\alpha} | \phi_{\beta} \rangle$$

- In an **orthonormal** basis, $\langle v | w \rangle = \sum_i v_i^* w_i$. In *any other* basis a matrix sits in the middle: $\langle v | w \rangle = \sum_{ij} v_i^* G_{ij} w_j$
- So G is just the table of inner products of the basis you happened to use — and $G = I$ *means* that basis is orthonormal
- Minimising $\frac{x^{\dagger} A x}{x^{\dagger} x}$ gives $Ax = \lambda x$; minimising $\frac{x^{\dagger} A x}{x^{\dagger} G x}$ gives $Ax = \lambda Gx$ — the Hc = ESc of Section 1.15
- Canonical form makes the block states orthonormal $\Rightarrow G = I$, an **ordinary** eigenproblem again

Why that matters in practice

Measured

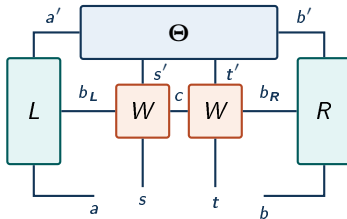
Random $L = 6$, $\chi = 4$ MPS: $\text{cond}(G) = 9.0 \times 10^9$. *The same state*, canonicalised: $\text{cond}(G) = 1$ and $\max |G - I| = 3.3 \times 10^{-16}$. Nothing physical changed — only the gauge.

- A generalised solver must factorise or solve with G — **never form G^{-1}** — and ill-conditioning amplifies rounding noise by $\text{cond}(G)$
- The gauge freedom $A^s \rightarrow A^s X$ lets block states become nearly *parallel*. Make two agree to one part in 10^6 : $\text{cond}(G) = 1.4 \times 10^{16}$ — numerically singular
- **A redundant bond dimension is not just wasteful, it is ill-conditioned**

Section 1.13 warns about exactly this for near-linearly-dependent basis sets. There the cure costs a diagonalisation; here it costs a QR.

Environments: freeze everything but two sites

$$\langle \Psi | \hat{H} | \Psi \rangle = \sum \Theta_{astb}^* L_{ab_L a'} W_{b_L c}^{ss'} W_{cb_R}^{tt'} R_{bb_R b'} \Theta_{a' s' t' b'}$$



Why three indices? A cut severs exactly three lines: one in the bra, one in the MPO, one in the ket.
So L, R are $\chi \times D \times \chi$.

Reading that diagram

- It is $\hat{H}_{\text{eff}} \Theta$, *not* $\langle \Psi | \hat{H} | \Psi \rangle$: the bra Θ^* has been removed
- **Primed** indices a', s', t', b' are summed against the input; the four **unprimed** legs dangle — they are the result
- Four legs in, four legs out, same shape (χ, d, d, χ) : that *is* “a linear map on the space Θ lives in”
- Θ is **one** box with *no internal bond*. The bond is created later, by the SVD

Never form $\hat{H}_{\text{eff}}$ as a matrix

Lanczos needs only the *action*: four contractions, $\mathcal{O}(\chi^3 d^2 D + \chi^2 d^3 D^2)$. At $\chi = 64$, $d = 4$ the matrix would be $65\,536 \times 65\,536$.

Why χ^3

$$\sum_{a'} L_{a b a'} \Theta_{a' s_1 s_2 b_r} \implies \boxed{\text{two free bond legs} \times \text{one summed bond leg} = \chi^3}$$

- Every object carries at most *two* bond legs: Θ is (χ, d, d, χ) , L and R are (χ, D, χ)
- The result keeps χ^2 entries, each costing a sum over χ terms
- This is just the cost of multiplying two $\chi \times \chi$ matrices — and that is every step
- Form an intermediate with **three** bond legs and you pay χ^4

The two-site update

1. **Form Θ** from the two central tensors. *The shared bond disappears, so the SVD may give it a new dimension — up to χd , not unlimited*
2. **Solve $\hat{H}_{\text{eff}} \Theta = E \Theta$** by Lanczos, matrix-free, from the current Θ . *The only place physics enters*
3. **Reshape and SVD:** $\Theta_{(as),(tb)} = U \Sigma V^\dagger$. *This is the Schmidt decomposition of that bond*
4. **Truncate** to χ ; record $\varepsilon_\chi = \sum_{k>\chi} \sigma_k^2$. *Optimal, by Eckart–Young*
5. **Move the centre** and extend the environment

Do not overconverge Lanczos in the early sweeps — the tensor is recomputed next sweep anyway — and tighten its tolerance near convergence. Converge the **sweep**, not the step.

How the centre actually moves

You never move Θ . You **split** it and keep one half.

$$\Theta_{astb} = \sum_c (M_i)_{asc} (B_{i+1})_{ctb} \xrightarrow{\text{SVD}} (A_i)_{asc} = U_{(as),c}, \quad (M_{i+1})_{ctb} = (\Sigma V^\dagger)_{c,(tb)}$$

$$\text{next step: } \Theta'_{ctub'} = \sum_b (M_{i+1})_{ctb} (B_{i+2})_{bub'}$$

direction	site i	site $i + 1$	extend
right	$A_i = U$ (left-can.)	$M_{i+1} = \Sigma V^\dagger$ (centre)	L_{i+1}
left	$M_i = U \Sigma$ (centre)	$B_{i+1} = V^\dagger$ (right-can.)	R_{i+1}

Σ always stays with the centre — it is the part that is *not* canonical.

One sweep, watched: there *and* back

moving right	1	2	3	4	5	6
start (all right-canonical)	R	R	R	R	R	R
bond (1,2)	L	C	R	R	R	R
bond (2,3)	L	L	C	R	R	R
bond (3,4)	L	L	L	C	R	R
bond (4,5)	L	L	L	L	C	R
bond (5,6)	L	L	L	L	L	L

then back	1	2	3	4	5	6
bond (5,6)	L	L	L	L	C	R
bond (4,5)	L	L	L	C	R	R
bond (3,4)	L	L	C	R	R	R
bond (2,3)	L	C	R	R	R	R
bond (1,2)	R	R	R	R	R	R
<i>ready for the next sweep</i>						

- C marches one site per step; an L wall grows behind it, the R wall retreats in front
- One sweep is the round trip: the return pass revisits every bond now that the *opposite* environment has moved. Canonical form, and so $G = I$, holds in either direction
- At the turn (5,6) is solved twice with *identical* environments, so an exact solve returns the *same state* (measured: overlap 1.000000000000, $\Delta E = 0$) — a *gauge move*, not an improvement. Hence $2(L - 1)$ local solves with this convention, $2L - 3$ if you skip it
- Beware: many papers call one *pass* a sweep, and the round trip two — a factor of two in “converged after 5 sweeps”

Pause to think: one site or two?

Pause to think #2

(about 4 min – discuss with your neighbour)

1. A one-site update optimises M_i alone, then moves on. Why can it *never* increase the bond dimension?
2. The two-site Θ is $\chi d \times d \chi$. What does its SVD let you do that the one-site version cannot?
3. What do you pay for it?

Answer to pause #2

1. The local eigenproblem lives in the space the *current* bonds already span. It re-mixes those χ states; it cannot invent new ones — and it is stuck in the symmetry sector it started in.
2. Its SVD has up to χd singular values, so it can **grow** χ exactly where the state needs it.
3. A factor d in cost, and a truncation at every step.

Single-site DMRG: the objection, and why it is not the whole story

The answer above is the classic objection — and it applies to the *plain* single-site update. With a **subspace expansion** it does not, and production codes often prefer single-site.

In favour of single-site

- Cheaper: no factor d from the two-site tensor, smaller SVD
- **No truncation step at all** — so the energy is *strictly* monotone
- Symmetry quantum numbers on the bond preserved exactly
- **With the expansion it can converge faster than two-site** while avoiding the same local-minimum traps: the expansion is *Hamiltonian-informed*

Against

- Plain version cannot grow χ , and sticks in its symmetry sector
- DMRG3S-type fixes need a mixing parameter α , annealed to zero (newer controlled bond expansion does not)
- No ε_χ for free — the error indicator must come from elsewhere

The mixer, and which to use

The cure for getting stuck is the **mixer** — a *subspace expansion* that perturbs the SVD so that the retained Schmidt states include directions the current state does not occupy yet.

In practice

Single-site stays *by construction in the subspace of a given bond-dimension MPS, unless you use a mixer* — so always run one. Two-site *formally avoids this problem* but can still get stuck: keep the mixer on for the *initial* sweeps, then switch it off to converge.

- So the ranking is: plain single-site *worst*; two-site safe in principle; single-site + expansion can be *best*
- With conserved quantum numbers two-site *can* create new sectors on the central bond — but not ones excluded by the **outer** bond spaces, and expansion helps when those are deficient

White (2005); Hubig, McCulloch & Schollwöck (2015), “strictly single-site” DMRG3S.

With two-site, the SVD makes the truncation — and its error indicator — visible.

The discarded weight is an error *indicator*

Write $|\psi_\chi\rangle = |\psi\rangle + |\delta\rangle$ with $\langle\delta|\delta\rangle = \varepsilon_\chi$. Since $|\psi\rangle$ is stationary, the first-order term vanishes:

$$E(\chi) - E_0 = \langle\delta|(\hat{H} - E_0)|\delta\rangle = \varepsilon_\chi \cdot \underbrace{\frac{\langle\delta|(\hat{H} - E_0)|\delta\rangle}{\langle\delta|\delta\rangle}}_{\langle\Delta E\rangle}$$

- The constant of proportionality is **an energy**: the mean excitation energy of what you threw away
- Second order in the state error, and always **positive** — variational

A trap to avoid

But ε_χ is the weight discarded at *one local SVD*, not $\|\delta\|^2$ against the exact ground state. The near-linear law is an empirical small- ε rule, **not a rigorous bound**, and other observables can be off by much more.

Run at several χ , plot E against ε_χ , extrapolate to $\varepsilon = 0$.

What the numbers look like

χ	E_{DMRG}	$E - E_0$	ε_χ	$S_{L/2}$
4	2.2435302961	4.1×10^{-1}	9.1×10^{-3}	1.0114
8	1.8655652393	3.6×10^{-2}	1.2×10^{-3}	1.0898
16	1.8326023377	3.5×10^{-3}	1.3×10^{-4}	1.1064
32	1.8291612166	5.5×10^{-5}	2.4×10^{-6}	1.1084
64	1.8291067166	4.5×10^{-8}	2.2×10^{-9}	1.1084
exact	1.8291066715			

pairing + particle-hole, $L = N = 8$, $g = 1$, $f = 0.5$. ε_χ here is the **maximum over bonds** in the last sweep — conventions differ, so say which you use.

- $\Delta E / \varepsilon_\chi$ drifts from 46 to 21 — **those numbers are energies**
- A line through the three *most accurate* runs lands within 5×10^{-6} ; through the three cheapest, it misses by 10^{-2}

Does the energy really fall monotonically?

Each update is *two* steps with opposite characters:

The eigensolve: cannot increase E

An exact minimisation over a subspace that *contains the current state*.

A trap to avoid

The truncation: *can*. Throwing away ε_χ moves the state and raises E by $\mathcal{O}(\varepsilon_\chi)$.

- Monotonicity is **not a theorem** — it is what happens once ε_χ is small
- Early on, with χ still growing, the energy can tick *upwards*. Not a bug
- What *is* guaranteed: every printed energy is an **upper bound** on E_0

How do you know it has converged?

Four things, all at once — and checking only the first is the commonest mistake:

1. the energy has stopped changing between sweeps
2. ε_χ is small and falls when you raise χ — flat energy + large ε_χ means not demonstrably converged: either stuck, or χ simply too small
3. $\langle \hat{N} \rangle$ is the value you asked for
4. $S_{L/2}$ has stopped growing with χ

Measured

How worried about local minima? For *these* models, not very: $L = 20$ Heisenberg from eight random starts agrees to 5×10^{-14} at $\chi = 4$. If your energies move with the seed, suspect a bug before suspecting physics. Genuine local minima live in frustration, all-to-all couplings and chemistry.

A 200-line NumPy code, Heisenberg at $L = 32$

χ	E_{DMRG}	ε_χ	$S_{L/2}$	time
8	-13.995308101464	4.0×10^{-5}	0.6941	0.3 s
16	-13.997257888143	3.1×10^{-6}	0.7203	0.5 s
32	-13.997315266845	1.2×10^{-8}	0.7215	0.9 s
64	-13.997315618007	6.9×10^{-12}	0.7215	2.1 s

- Energy falls monotonically, by one to two orders per doubling of χ
- $S_{L/2}$ has converged by $\chi = 32$ — which is why the last row buys so little
- Dense diagonalisation in the full 2^{32} basis is impractical; this takes two seconds

Report ε_χ as well as χ — and the sweep schedule and tolerances.

Where the name comes from: White, 1992

- Grow a block site by site; its space grows as d^ℓ , so truncate to χ states. Which χ ?
- Wilson's NRG kept the lowest eigenstates of the *block* Hamiltonian — wrong once the block is embedded
- White: form the *superblock*, get its ground state, trace out the environment, keep the largest eigenvectors of ρ_A

That is exactly our Schmidt truncation

The eigenvectors of ρ_A are the Schmidt states; its eigenvalues are σ_k^2 ; keeping the largest minimises the discarded weight. “Density matrix” = the reduced density matrix of the cut; “renormalisation group” = the block growing.

Infinite-system algorithm: grow to length L . *Finite-system* algorithm: then sweep — what we described.

Exercise session

Exercise 1: an MPS for the W state

$$|W\rangle = \frac{1}{\sqrt{3}}(|100\rangle + |010\rangle + |001\rangle)$$

1. Write the coefficient matrix for the cut after site 1 and find its rank. What is χ_1 ? Same for χ_2 .
2. Construct A_i^s explicitly. *Hint*: let the bond index record one bit — **has the excitation been placed yet?**
3. Check that $C_{100} = C_{010} = C_{001} = 1/\sqrt{3}$. Why does $C_{110} = 0$ come out automatically?
4. Generalise to L sites. Does χ grow with L ?
5. Compare with GHZ, $\frac{1}{\sqrt{2}}(|0\cdots 0\rangle + |1\cdots 1\rangle)$, which also has $\chi = 2$. In each case, *what does the bond index remember?*

Exercise 2(a): right- and mixed-canonical form

In the lectures of Week 40, we built the left-canonical MPS by $L - 1$ SVDs sweeping *rightwards*. Now do it the other way.

1. Repeat the construction starting from the *right* end. Show that the tensors you get satisfy

$$\sum_s B^s B^{s\dagger} = I$$

and that this makes the *right* block states orthonormal.

2. Now stop halfway: sweep in from the left to site i and in from the right to site $i + 1$. Show that

$$|\Psi\rangle = \sum_{\{s\}} A^{s_1} \dots A^{s_i} \Lambda_i B^{s_{i+1}} \dots B^{s_L} |s_1 \dots s_L\rangle$$

and that Λ_i is diagonal with the Schmidt values of bond i on it.

Exercise 2(b): the two rules that make overlaps cheap

$\langle \Psi | \Psi \rangle$ is a double-layer network. Prove the collapse rule, then use it.

$$\begin{array}{c} \begin{array}{|c|} \hline A \\ \hline \bar{A} \\ \hline \end{array} \quad = \quad \left(\quad \sum_s A^{s\dagger} A^s = I \quad (\text{left-canonical}) \right. \\ \begin{array}{|c|} \hline B \\ \hline \bar{B} \\ \hline \end{array} \quad = \quad \left. \quad \right) \quad \sum_s B^s B^{s\dagger} = I \quad (\text{right-canonical}) \end{array}$$

1. Prove each rule from $\sum_s A^{s\dagger} A^s = I$ and $\sum_s B^s B^{s\dagger} = I$.
2. Zip the network up from the left, one site at a time. Show that the contraction reduces to the norm of the final tensor — so an MPS is normalised exactly when *every* tensor, that one included, is left-isometric.
3. What is the cost of $\langle \Phi | \Psi \rangle$ for two *different* MPS, where nothing collapses? Why must you contract site by site and never form the d^L vector?

Exercise 2(c): overlaps from the orthogonality centre

Put the state in mixed-canonical form. Everything outside the centre now collapses by 2(b).

1. **Bond centre.** Show $\langle \Psi | \Psi \rangle = \|\Lambda_i\|_F^2 = \sum_a \sigma_a^2$, and that the entanglement entropy of that cut is $-\sum_a \sigma_a^2 \ln \sigma_a^2$ — both readable off the centre alone, with *no* contraction.
2. **Two-site centre.** With $\Theta^{s_i s_{i+1}} = A^{s_i} \Lambda_i B^{s_{i+1}}$, show $\langle \Psi | \Psi \rangle = \|\Theta\|_F^2$.

This is why the DMRG eigenproblem is an ordinary one: $G = I$ comes from exactly these two rules.

Exercise 3: how many parameters does an MPS really have?

An MPS with bond dimensions $\chi_0 = 1, \chi_1, \dots, \chi_L = 1$ is built from $\sum_{i=1}^L d \chi_{i-1} \chi_i$ numbers. But the gauge freedom $A_i^s \rightarrow A_i^s X_i, A_{i+1}^s \rightarrow X_i^{-1} A_{i+1}^s$ means many of them describe the *same* state.

1. At each internal bond X_i is any invertible $\chi_i \times \chi_i$ matrix. How many parameters does that remove?
2. Write down the count of *independent* parameters.
3. For the **maximal generic exact ranks**, $\chi_i = \min(d^i, d^{L-i})$, evaluate it for $d = 2$ and $L = 2, 3, 4$. What do you get?
4. Prove it in general — and explain why the answer *had* to be that, in one sentence.
5. Now cap χ *below* those generic ranks. Why is the count then strictly smaller, and what does that say about the ansatz?

Summary

Summary of week 41

Thursday: operators

- W is $D \times D$ with $d \times d$ entries; D counts terms open at a cut
- First row = start, last column = finish, **bulk** = **memory**
- Minimal D is the operator Schmidt rank — the automaton can overcount
- JW strings cost no D , but cost locality

Friday: the algorithm

- Canonical form $\Rightarrow G = I \Rightarrow$ ordinary eigenproblem
- L, R are $\chi \times D \times \chi$: a cut severs three lines
- Cost $\mathcal{O}(L\chi^3 d^2 D)$ — two free bond legs, one summed
- Split Θ , keep half, move on. SVD everywhere
- ε_χ is an error indicator; extrapolate with it

Answers to the week-40 exercises

Answers to the week-40 exercises: 1a, the norm

1a) The norm to second order

With $\hat{T} = \sum_{ai} \delta C_{ai} a_a^\dagger a_i$ and $\hat{T}_i^2 = 0$ (week 39, 2c), $e^{\hat{T}} = \prod_i (1 + \hat{T}_i) = 1 + \hat{T} + \sum_{i < j} \hat{T}_i \hat{T}_j + \dots$: the second-order piece is the sum of *products of two different* $\hat{T}_i$, a $2p-2h$ operator. In $\langle c' | c' \rangle$: 1 (zeroth order); $\langle c | \hat{T} | c \rangle = 0$ and $\langle c | \hat{T}_i \hat{T}_j | c \rangle = 0$ (odd and $2p-2h$ against the vacuum); $\langle c | \hat{T}^\dagger \hat{T} | c \rangle = \sum_{ai, bj} \delta C_{ai}^* \delta C_{bj} \langle \Phi_i^a | \Phi_j^b \rangle = \sum_{ai} |\delta C_{ai}|^2$. The cross term $\hat{T}^\dagger \hat{T}_i \hat{T}_j$ is a $1p-1h$ bra against a $2p-2h$ ket and vanishes, so the first correction to $1 + \sum |\delta C|^2$ is in fact of *fourth* order, $\| \sum_{i < j} \hat{T}_i \hat{T}_j | c \rangle \|^2$ – safely inside the $\mathcal{O}(\delta C^3)$ claimed.

- Rule used throughout: $\langle c | \hat{O} | c \rangle$ of a normal-ordered product with respect to $|c\rangle$ vanishes unless the product is empty – a quasi-particle creator anywhere kills the vacuum expectation value.
- The same expansion, continued to the energy, is what produces the three blocks of $\hat{M}$ in 1e.

Answers to the week-40 exercises: 1b, the one-body part

1b) $\langle c | \{a_i^\dagger a_a\} \hat{F}_N \{a_b^\dagger a_j\} | c \rangle$: two patterns

Number the operators $a_i^\dagger(1) a_a(2) a_p^\dagger(3) a_q(4) a_b^\dagger(5) a_j(6)$, with $\hat{F}_N = \sum_{pq} f_{pq} \{a_p^\dagger a_q\}$. Pairs inside one $\{ \}$ are never contracted, and a non-zero contraction needs a particle pair $a_\alpha a_\beta^\dagger$ ($\alpha, \beta > F$) or a hole pair $a_\alpha^\dagger a_\beta$ ($\alpha, \beta \leq F$), the first factor to the left. That leaves exactly two complete patterns:

pattern	contractions	crossings	value
I	$(2, 3) (4, 5) (1, 6): \delta_{ap} \delta_{qb} \delta_{ij}$	0, sign +	$+f_{ab} \delta_{ij}$
II	$(1, 4) (2, 5) (3, 6): \delta_{iq} \delta_{ab} \delta_{pj}$	3, sign -	$-f_{ji} \delta_{ab}$

Sum: $f_{ab} \delta_{ij} - f_{ji} \delta_{ab}$. In the Hartree-Fock basis $f_{pq} = \varepsilon_p \delta_{pq}$, so this is $(\varepsilon_a - \varepsilon_i) \delta_{ab} \delta_{ij} = \Delta_{ai, bj}$: the one-body part of the stability matrix is the diagonal of particle-hole energies, exactly as in week 39.

Answers to the week-40 exercises: 1c, the particle-hole interaction

$$1c) \langle c | \{ a_i^\dagger a_a \} \hat{V}_N \{ a_b^\dagger a_j \} | c \rangle = A_{ai,bj}$$

$\hat{V}_N = \frac{1}{4} \sum_{pqrs} \langle pq | \hat{v} | rs \rangle_{AS} \{ a_p^\dagger a_q^\dagger a_s a_r \}$; number $a_i^\dagger(1) a_a(2) a_p^\dagger(3) a_q^\dagger(4) a_s(5) a_r(6) a_b^\dagger(7) a_j(8)$. a_a must meet $a_p^\dagger$ or $a_q^\dagger$, $a_b^\dagger$ must meet a_s or a_r , $a_i^\dagger$ the other of a_s, a_r , a_j the other of $a_p^\dagger, a_q^\dagger$: $2 \times 2 = 4$ patterns. Pattern $(2,3)(4,8)(1,5)(6,7)$, i.e. $p=a, q=j, s=i, r=b$: the only crossing is $(1,5)$ with $(4,8)$, sign $-$, value $-\frac{1}{4} \langle aj | \hat{v} | bi \rangle_{AS}$. Swapping $r \leftrightarrow s$ adds one crossing *and* one antisymmetry sign; swapping $p \leftrightarrow q$ likewise. All four patterns are equal, and

$$A_{ai,bj} = 4 \times \left(-\frac{1}{4} \langle aj | \hat{v} | bi \rangle_{AS} \right) = -\langle aj | \hat{v} | bi \rangle_{AS} = \langle aj | \hat{v} | ib \rangle_{AS}.$$

The same number is $\langle \Phi_i^a | \hat{V}_N | \Phi_j^b \rangle$, the matrix the Tamm-Dancoff approximation diagonalises (chapter 7).

- Sign rule: $(-1)^{\text{number of crossings}}$ of the contraction lines, with every line drawn above the operators; the two swaps each change the count by one *and* the matrix element by a sign, which is why all four patterns agree.
- A is the *exchange-like* coupling of one particle-hole pair to another; it is what distinguishes TDA from a sum of independent excitations.

Answers to the week-40 exercises: 1d, the pair-creation part

$$1d) \langle c | \hat{V}_N a_a^\dagger a_i a_b^\dagger a_j | c \rangle = B_{ai,bj}^*$$

Number $a_p^\dagger(1) a_q^\dagger(2) a_s(3) a_r(4) a_a^\dagger(5) a_i(6) a_b^\dagger(7) a_j(8)$. Contractions among (5)–(8) vanish (a against i is neither a particle nor a hole pair), so a_s, a_r pair with $a_a^\dagger, a_b^\dagger$ and $a_p^\dagger, a_q^\dagger$ with a_i, a_j : four patterns again. Pattern (3, 5)(4, 7)(1, 6)(2, 8), $s = a, r = b, p = i, q = j$: crossings (3, 5) $\times$ (4, 7), (4, 7) $\times$ (1, 6), (1, 6) $\times$ (2, 8) – three, sign $-$, value $-\frac{1}{4} \langle ij | \hat{v} | ba \rangle = +\frac{1}{4} \langle ij | \hat{v} | ab \rangle_{AS}$; all four agree, total $\langle ij | \hat{v} | ab \rangle_{AS} = \langle ab | \hat{v} | ij \rangle_{AS}^* = B_{ai,bj}^*$. Check without Wick: $a_a^\dagger a_i a_b^\dagger a_j | c \rangle = a_a^\dagger a_b^\dagger a_j a_i | c \rangle = |\Phi_{ij}^{ab}\rangle$, and $\langle c | \hat{V}_N | \Phi_{ij}^{ab} \rangle = \langle ij | \hat{v} | ab \rangle_{AS}$ is the Condon-Slater rule. E_0 gives $E_0 \langle c | \Phi_{ij}^{ab} \rangle = 0$; $\hat{F}_N$, a one-body operator, cannot bridge a double excitation. Pauli: $a_a^\dagger a_a^\dagger = 0$ and $a_i a_i = 0$, so the element vanishes unless $a \neq b$ and $i \neq j$ – which is also what the antisymmetry of $\langle ij | \hat{v} | ab \rangle_{AS}$ says.

Answers to the week-40 exercises: 1e, assembling the stability matrix

$$1e) \Delta E = \frac{1}{2} \langle \chi | \hat{M} | \chi \rangle$$

Expand $E = \langle c' | \hat{H} | c' \rangle / \langle c' | c' \rangle$ to second order with $e^{\hat{T}} = 1 + \hat{T} + \frac{1}{2} \hat{T}^2 + \dots$ and $\hat{H} = E_0 + \hat{F}_N + \hat{V}_N$:

$$\langle c' | \hat{H} | c' \rangle = E_0 \left(1 + \sum |\delta C|^2 \right) + \sum \delta C_{ai}^* \delta C_{bj} (\Delta + A)_{ai,bj} + \frac{1}{2} \sum \delta C_{ai} \delta C_{bj} B_{ai,bj}^* + \frac{1}{2} \sum \delta C_{ai}^* \delta C_{bj} B_{ai,bj} + \mathcal{O}(\delta C^3),$$

the middle term from (b) and (c), the last two from (d) and its conjugate ($\frac{1}{2} \hat{T}^2$ carries the $\frac{1}{2}$). Dividing by the norm of (a) and subtracting E_0 leaves

$$\Delta E = \frac{1}{2} \begin{pmatrix} \delta C^* & \delta C \end{pmatrix} \begin{pmatrix} \Delta + A & B \\ B^* & (\Delta + A)^* \end{pmatrix} \begin{pmatrix} \delta C \\ \delta C^* \end{pmatrix} = \frac{1}{2} \langle \chi | \hat{M} | \chi \rangle.$$

Hermitian: $(\Delta + A)_{bj,ai} = \langle bi | \hat{v} | ja \rangle = \langle aj | \hat{v} | ib \rangle^* = (\Delta + A)_{ai,bj}^*$ and $B_{bj,ai} = \langle ba | \hat{v} | ji \rangle = B_{ai,bj}$, so $B^T = B$, the lower-left block is $B^* = B^\dagger$, and $\hat{M}^\dagger = \hat{M}$.

Answers to the week-40 exercises: 1e, one pair, and what the diagonal means

One pair

Keep a single (a, i) . Then $B_{ai,ai} = \langle aa|\hat{v}|ii\rangle_{AS} = 0$ by antisymmetry, $\hat{M} = \text{diag}(\Delta + A, \Delta + A)$ and

$$\Delta E = |\delta C_{ai}|^2 (\epsilon_a - \epsilon_i + A_{ai,ai}) = |\delta C_{ai}|^2 (\epsilon_a - \epsilon_i - \langle ai|\hat{v}|ai\rangle_{AS}),$$

since $A_{ai,ai} = \langle ai|\hat{v}|ia\rangle = -\langle ai|\hat{v}|ai\rangle$. This is the curvature $\langle \Phi_i^a|\hat{H}|\Phi_i^a\rangle - E_0$ of the parabola of week 39 (and of week-39 answer 1d): the diagonal of the stability matrix is the energy of a single particle-hole excitation *including* its interaction with the hole it left behind.

Answers to the week-40 exercises: 2a–b, what “positive semidefinite” means

2a) Three statements, and which one is “stable in every direction”

Spectral theorem: $\hat{M} = \sum_k \lambda_k |k\rangle\langle k|$ with orthonormal $|k\rangle$, so $\langle \chi | \hat{M} | \chi \rangle = \sum_k \lambda_k |\langle k | \chi \rangle|^2$. All $\lambda_k \geq 0$ makes this ≥ 0 ; conversely $\chi = |k\rangle$ gives $\lambda_k \geq 0$. If $\hat{M} = C^\dagger C$ then $\langle \chi | \hat{M} | \chi \rangle = \|C\chi\|^2 \geq 0$ (and conversely $C = \hat{M}^{1/2}$ works). “ $\Delta E \geq 0$ in every direction” is the quadratic-form statement, $\frac{1}{2} \langle \chi | \hat{M} | \chi \rangle \geq 0$ for every $\chi = (\delta C, \delta C^*)$.

Those χ are not all vectors – but the block structure of $\hat{M}$ means that if (u, v) is an eigenvector with eigenvalue λ so is (v^*, u^*) , and their sum has the form $(\delta C, \delta C^*)$; so the smallest value of the form on physical directions is $\lambda_{\min}(\hat{M})$, and the two statements agree.

2b) Why leading minors are not enough, and the hierarchy of necessary conditions

$\text{diag}(0, -1)$: leading minors $M_{11} = 0$ and $\det = 0$, both ≥ 0 ; eigenvalue -1 . The *non-leading* principal minor $M_{22} = -1$ exposes it – Sylvester’s test with $\geq$ fails because a zero on the diagonal hides whatever follows. Restricting χ to $(\delta C, 0)$ gives $\langle \chi | \hat{M} | \chi \rangle = \delta C^\dagger (\Delta + A) \delta C$: $\Delta + A$ is the principal submatrix of $\hat{M}$ on the first half of the indices, and a principal submatrix of a positive semidefinite matrix is positive semidefinite (take χ supported on those indices). Necessary; not sufficient, since a general χ also feels B . The diagonal condition $\varepsilon_a - \varepsilon_j + A_{ai, ai} \geq 0$ is the 1×1 principal submatrices of $\Delta + A$: weaker again – $\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$ has a positive diagonal and eigenvalue -1 .

Answers to the week-40 exercises: 2c, the Lipkin conditions one by one

2c) Lipkin, $N = 2$, $W = 0$: the conditions, one by one

Order the components $(\delta C_1, \delta C_2, \delta C_1^*, \delta C_2^*)$:

$$\hat{M} = \begin{pmatrix} \varepsilon & 0 & 0 & V \\ 0 & \varepsilon & V & 0 \\ 0 & V & \varepsilon & 0 \\ V & 0 & 0 & \varepsilon \end{pmatrix}, \quad \text{eigenvalues } \varepsilon \pm V, \text{ each twice.}$$

Diagonal: $\varepsilon > 0$, always. Trace: $4\varepsilon > 0$, always. 2×2 principal minors: ε^2 within a block, and $\varepsilon^2 - V^2$ for the pairs (1, 4) and (2, 3) – **the first to fail**, at $|V| = \varepsilon$. 3×3 : $\varepsilon(\varepsilon^2 - V^2)$, fails at the same point. Determinant: $(\varepsilon^2 - V^2)^2 \geq 0$ *always* – the two negative eigenvalues beyond $|V| = \varepsilon$ multiply to a positive number, so $\det \geq 0$ never notices. Sylvester's *leading* minors $\varepsilon, \varepsilon^2, \varepsilon(\varepsilon^2 - V^2), (\varepsilon^2 - V^2)^2$: the third fails. All of this is $\lambda_{\min} = \varepsilon - |V| = \varepsilon(1 - \chi)$ with $\chi = |V|(N - 1)/\varepsilon = |V|/\varepsilon$: the Hartree-Fock reference becomes unstable exactly at $\chi = 1$, where week 39 found the deformed solution taking over.

Answers to the week-40 exercises: 2d, the pairing block

2d) Pairing: $\begin{pmatrix} \Delta & -g/2 \\ -g/2 & \Delta \end{pmatrix}$, $\Delta = \varepsilon_q - \varepsilon_p + g/2$

Minor: $\Delta^2 - g^2/4 = (\Delta - g/2)(\Delta + g/2) = (\varepsilon_q - \varepsilon_p)(\varepsilon_q - \varepsilon_p + g) \geq 0$ for $g \geq 0$. Eigenvalues $\Delta \mp g/2$: $\varepsilon_q - \varepsilon_p$ on $(1, 1)/\sqrt{2}$ and $\varepsilon_q - \varepsilon_p + g$ on $(1, -1)/\sqrt{2}$. The lower eigenvector has $\delta C_{q\uparrow, p\uparrow} = \delta C_{q\downarrow, p\downarrow}^*$: both members of the pair move *together*, the state stays in seniority zero, and the diagonal pairing energy of a seniority-zero determinant, $-\frac{g}{2} \times (\text{number of pairs})$, does not depend on *which* levels the pairs sit on – so the interaction cancels and only the level spacing is paid. Breaking a pair coherently costs nothing at the mean-field level beyond $\varepsilon_q - \varepsilon_p$: the determinant is stable for every g , and the pairing correlation (pair scattering, a $2p$ - $2h$ effect) is invisible to it. Hence BCS in chapter 7, and the bond dimension of this week.

Answers to the week-40 exercises: 2e, when $\Delta + A$ itself fails

2e) A negative eigenvalue of $\Delta + A$

$\Delta + A$ is the matrix of $\hat{H} - E_0$ in the space of single excitations $|\Phi_i^a\rangle$ – the Tamm-Dancoff (CIS) matrix of chapter 7. Its eigenvector u with $\lambda < 0$ is a $1p\text{-}1h$ state $\sum_{ai} u_{ai} |\Phi_i^a\rangle$ whose energy lies **below the reference**: a CIS calculation on this $|c\rangle$ returns a negative “excitation energy”. On the manifold, take $\delta C = e^{i\varphi} u$ with φ chosen so that $\text{Re}(e^{2i\varphi} u^T B^* u) \leq 0$; then $\Delta E \leq \lambda \|u\|^2 < 0$, so $\hat{M} \not\geq 0$ whatever B is.

Why this is the worse case

- When only B breaks stability, $\Delta + A \geq 0$: every single excitation costs energy, CIS/TDA is healthy, and the trouble shows up only in the RPA, whose frequencies ω^2 turn negative – the instability lives in the *coupling* of excitation to de-excitation, i.e. in ground-state correlations (the Lipkin case above: $A = 0$, all of it in B).
- When $\Delta + A$ fails, the reference is not even the lowest state of the space of single excitations: a *single* orbital rotation lowers the energy, the Hartree-Fock iteration has converged to a saddle, and both TDA and RPA on it are meaningless. The cure is to follow u downhill to the deformed determinant, as for Lipkin at $\chi > 1$ – after which a new $\hat{M}$ must be checked.
- Thouless’ theorem on the RPA (chapter 7): the RPA equations have real frequencies if and only if $\hat{M} \geq 0$. The two cases are therefore “RPA complains” versus “even TDA complains”.

Next week, and a closing remark

Week 42, October 15–16: back to the mean field – the electron gas, and density functional theory

Thursday: the infinite homogeneous electron gas in Hartree-Fock (chapter 6, section 6.14): plane waves are the self-consistent orbitals, the direct term cancels against the background, the exchange energy in closed form, the single-particle energies and their logarithmic singularity at k_F , the energy per particle in r_s and the Wigner-Seitz radius.

Friday: density functional theory (chapter 8): the electronic Hamiltonian and reduced density matrices, the Hohenberg-Kohn theorems, the Kohn-Sham construction, and the local density approximation built from the electron gas of Thursday.

Reading: chapter 6, section 6.14; chapter 8, sections 8.1–8.6; Szabo-Ostlund, chapter 3 for the plane-wave Hartree-Fock. The exercises 1–3 of this week are answered on the week-42 slides.

Two weeks of tensor networks have given us a method with no reference determinant at all: the singular value decomposition at every bond, an automaton for the Hamiltonian, and a sweep of small eigenproblems whose error we can read off the discarded weight. Next week we return to the determinant – and to the one system where the Hartree-Fock equations can be solved by hand.

From entanglement back to the mean field: the electron gas, and the density as the variable.